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Enterprise Portfolio Optimization Based on Particle Swarm Optimization and Quantum Approximate Optimization Algorithm

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***Abstract*—With the development of intelligent algorithms, new challenges have emerged in the financial field in optimizing the investment portfolio. In view of the current problems existing in portfolio optimization, the particle swarm algorithm (Particle Swarm Optimization, PSO) and quantum approximation optimization algorithm (Quantum Approximate Optimization Algorithm, QAOA) of the initial population Qanta coding is improved, and a hybrid algorithm suitable for solving enterprise portfolio optimization problems is proposed. The results found that the average performance of this method is below 0.06, which is better compared with the comparison algorithm. At the same time, with the increase of the risk level, the maximum, minimum and average increase of the yield are 10.99%, 19.80% and 7.98% respectively, stabilizing. The results show that the research algorithm is effective in solving the enterprise portfolio optimization problem, has better optimization ability in practical application, and can provide reference for enterprise investors to make investment decisions.**

Keywords—Particle Swarm Optimization, quantum approximate optimization algorithm, portfolio, intelligent algorithm, risk, return

I. Introduction

With the continuous development of artificial intelligence technology, and the intersection and penetration with other disciplines, especially in the field of enterprise portfolio optimization, its importance is increasingly prominent. Portfolio optimization is a process of rationally allocating investment resources according to the established target returns and allowable risks to diversify risks and improve returns [1-2]. Under the continuous research of scholars, the portfolio model has gradually developed into semi-variance model, mean absolute deviation model, capital pricing and arbitrage pricing model, etc [3-4]. Portfolio optimization model usually has complex nonlinear structure, ultra-high dimensional decision variables and extremely wide decision space, which makes this problem a difficult NP problem, so it is often difficult to solve [5] with traditional computational methods. In view of this problem, the research innovatively combines the optimization of particle swarm algorithm (Particle Swarm Optimization, PSO) and quantum approximation optimization algorithm (Quantum Approximate Optimization Algorithm, QAOA) to solve enterprise portfolio, in order to help enterprise investors to obtain portfolio optimization results more quickly and provide strong support for investment decisions.

II. Research Foundation

With the increasing maturity of quantum computing technology, it began to be widely used in the portfolio optimization problem with its efficient solution efficiency. For the portfolio optimization problem, Brandhofer et al. [6] tested the portfolio optimization performance with QAOA, defining a standard that distinguishes between "simple" and "difficult" examples of portfolio optimization problems. In order to solve the problems related to portfolio optimization, Hegade et al. [7] proposed a portfolio optimization method based on digital anti-antiadiabatic quantum algorithm. The results find that when the approximate antiadiabatic technology is introduced, the success probability of the resulting digital quantum algorithm is significantly improved. For the portfolio optimization problem with specific objectives or constraints, Aguilera et al. proposed a multi-target portfolio optimization method based on quantum annealing algorithm, which found that the potential [8] method provides more effective and powerful solutions for portfolio management. In order to solve the portfolio problem that is difficult to solve by classical algorithms, Qu et al. [9] proposed a portfolio optimization method based on quantum walk. The results find that by increasing the number of phase shift and quantum walk iterations, the probability of the state corresponding to the high-quality solution is effectively amplified, which can provide a reference for financial portfolio optimization. To analyze the efficiency of the car company portfolio, using the PSO and Empire competition algorithm, Hamd et al. [10] found that the proposed portfolio optimization model shows that the method shows good performance. To solve the problem of multi-target optimization in an investment portfolio, Erwin et al. [11] proposed a multi-target portfolio optimization algorithm. Results show that the method has similar performance to other multi-objective algorithms while obtaining a more diverse set of optimal solutions. To sum up, in the field of enterprise portfolio optimization problem, the research on intelligent solving algorithm has achieved some results. In particular, the application of quantum computing to portfolio optimization has become a current research hotspot. However, the current research on the portfolio optimization problem still has the limitations of rational person hypothesis and risk attitude hypothesis. Based on this, the research innovatively combines PSO and QAOA algorithm to propose a hybrid algorithm suitable for solving the mathematical model of enterprise portfolio optimization, so as to help enterprises arrange investment reasonably and formulate accurate investment scale.

III. Research Methods

A. Construction of the Mathematical Model of the Enterprise Portfolio Optimization Problem

In the process of pursuing portfolio optimization, the risks faced by investors mainly stem from a variety of

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uncertain factors, which may lead to the deviation between their actual return and their expected return [12]. In practice, the investment result is high return with high risk, while low risk usually brings low return. Markowitz The proposed portfolio mean-variance theory plays a central role in this field. But its application is based on a key assumption: the asset returns follow a normal distribution, and it is difficult to hold up in real market operations [13-14]. In this regard, a comprehensive measure of income and risk is proposed to build a mathematical model of enterprise portfolio optimization problem. In Markovwitz's classic mean-variance portfolio optimization framework, investment return is assessed by calculating the expected return rate of the portfolio, while investment risk is measured by the variance of the return rate. The mathematical model is formula (1).

$$\min R = \sum_{i=1}^{M}\sum_{j=1}^{M} x_i x_j \operatorname{cov}\left(r_i, r_j\right) \tag{1}$$

In equation (1), x_j represents the investment share of the securities, r_j represents the average return rate of the securities, M represents the number of investment risk securities, and $\operatorname{cov}\left(r_i, r_j\right)$ represents the covariance of the return rate of the securities. The constraints of the mathematical model are given in equation (2).

$$\begin{cases} P = \sum_{j=1}^{M} x_i x_j \geq P* \\ \sum_{j=1}^{M} x_j = 1, x_j \geq 0 \end{cases} \tag{2}$$

In Equation (2), $P*$ represents the expected rate of return, and the Markovwitz mean variance model requires the minimum investment risk on the premise of meeting the expected rate of return. Of course, the model can also be expressed as the maximum investment return at a given risk level, see formula (3).

$$\max P = \sum_{j=1}^{M} x_i x_j \tag{3}$$

In the discussion of risk securities portfolio, return and risk are mutual cost and compensation[15]. In order to balance both risk and benefit, the research integrates them into a comprehensive evaluation index. Given that risk is the primary consideration of investors and the risk that investors can bear is usually limited in their portfolio, the risk is given greater weight when constructing this comprehensive index. As the gain gets larger and the composite metric converges to the optimal value, the study calls it the return Risk Evaluation Criterion (Return Risk Criterion, PRC), see formula (4).

$$F\left(P, R\right) = \frac{P}{1 + \gamma\left(R\right).e^{\lambda\left(R - R*\right)}} \tag{4}$$

In equation (4), λ represents the amplification coefficient, R represents the actual risk level, P represents the return, and $R*$ represents the risk tolerance level of the investor. In the iterative process of the algorithm, the core function of PRC is to ensure that a balance between the actual risk and the return is maintained within the investor's risk tolerance. The fundamental reason for adopting this asymmetric criterion is that if the actual risk is beyond the investor, a large penalty is required for this situation. Only when the actual risk remains within the preset risk level will the benefits be amplified.

B. Mathematical Model Solution of Enterprise Portfolio Optimization Problem Based on PSO and QAOA Algorithm

After constructing the mathematical model of enterprise portfolio optimization problem, the paper combines PSO and QAOA algorithm. First, the study assumes that the quantum population contains N individuals, the risk securities type of investment is M, the population of N individuals is $X(t)$, each individual $x_i(t)$ represents N solution of the problem, that is, the dimension of each individual is M. To ensure the convergence of the properties of the algorithm, each individual converges to its local attractor $P_i(t)$, and the j-dimensional definition of the local attractor $P_i(t)$ is described in the formula (5).

$$P_{ij}(t) = v.u_{ij}(t) + (1 - v).g_j(t), v: \ U\left[0,1\right] \tag{5}$$

In Equation (5), $u_i(t)$ represents the optimal position of individual i so far, which is called the local optimal solution, and $g_j(t)$ is the optimal position in the whole population so far, which is called the global optimal solution. Each individual is in a potential well in every dimension during the iteration, assuming that the length of the potential well is $L_{ij}\left(t\right)$, which determines the search range of each individual, where t represents the number of iterations. For the wave function expression of each individual state in quantum space, see formula (6).

$$\delta\left(x_{ij}(t+1)\right) = \frac{1}{\sqrt{L_{ij}\left(t\right)}} e^{-\left|P_{ij}(t) - x_{ij}(t+1)\right| / L_{ij}(t)} \tag{6}$$

In Equation (6), $x_{ij}(t+1)$ represents the j dimension of individual $x_i(t+1)$ in generation $t+1$, and $P_{ij}(t)$ is the j dimension of the local attractor $P_i(t)$, from which the corresponding probability density function H is obtained, see Equation (7).

$$H\left(x_{ij}(t+1)\right) = \frac{1}{L_{ij}\left(t\right)} e^{-2\left|P_{ij}(t) - x_{ij}(t+1)\right| / L_{ij}(t)} \tag{7}$$

After updating the position of the individual, in order to maintain the feature length, the concept $m_j\left(t\right)$ introduces the mean optimal solution of j population, which is

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expressed as the average of the local optimal solution of all particles, see formula (8).

$$\begin{cases} m_j(t) = \dfrac{1}{N}\sum_{i=1}^{N} u_{ij}(t) \\ L_{ij}(t) = 21\left|m_j(t) - x_{ij}(t)\right| \end{cases} \tag{8}$$

In the equation (8), 1 represents the contraction and expansion coefficient, which is used to control the convergence rate of the algorithm. Thus, the calculation formula of the new position of each individual is shown in the formula (9).

$$x_{ij}(t+1) = P_{ij}(t) \pm 1\left|m_j(t) - x_{ij}(t)\right| \ln\frac{1}{u} \tag{9}$$

Finally, the mathematical model of enterprise portfolio optimization problem based on PSO and QAOA algorithm was solved, as shown in Fig. 1.

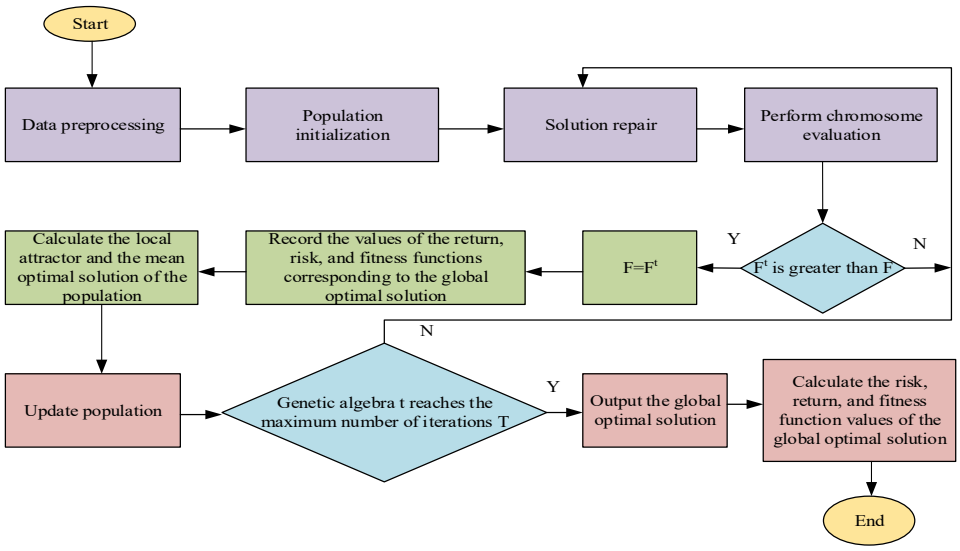

Fig. 1. Flow chart of solving the mathematical model of the enterprise portfolio optimization problem

In Fig. 1, first, import the yield of M risk securities in n trading days, and the average yield and covariance matrix are calculated for each of the risk securities. Then the initial population is randomly generated, requiring the individual to meet the total constraint as feasible solution, otherwise not feasible solution. To improve the validity of the search, the unfeasible solution is repaired so that it satisfies the total constraint, recording the repaired population as $X(t)$. At the same time, the optimal fitness and the corresponding chromosome and feasible solution of the t-generation population were screened out. Then, the optimal fitness in generation t is compared with the optimal fitness size of the current population, and the values of the benefits, risks and fitness functions corresponding to the global optimal solution are recorded. After elite selection, the mean best solution of the population. Finally, the position of each chromosome is updated to output the global optimal solution if the number of genetic generations reaches the maximum number of iterations.

IV. Application Effect Analysis

In order to verify the proposed enterprise portfolio optimization method, 20 stocks in different sectors of a stock exchange were selected, and the yield data of 50 trading days from June 24,202 4 to September 1,202 4 were selected for analysis. The simulation parameters and the experimental operating environment setting are shown in Table I.

TABLE I. Simulation Parameters and Experimental Environment Settings

/	parameter	numeric value
Simulation parameters	Population size	20
	Maximum iteration times	200
	The magnification factor in the income risk evaluation criteria	50
	Investment lower limit	0.05
	Investment ceiling	0.3
	The encoding length of the quantum chromosome	10*14
Experimental running environment	programming	MatlabR2018b
	Computing platform	Intel(R)Core(TM)i7-4702MQCPU@2.20GHz Processor
	memory	RAM4GB

To verify the effectiveness of the proposed method, the proposed algorithm is first compared with the differential evolution algorithm (Differential Evolution, DE), genetic algorithm (Genetic Algorithm, GA) and traditional PSO. The historical average of the 20 stocks was used as their expected return rate, and the average and standard deviation of the risks under different expected returns were calculated, and the results are shown in Fig. 2.

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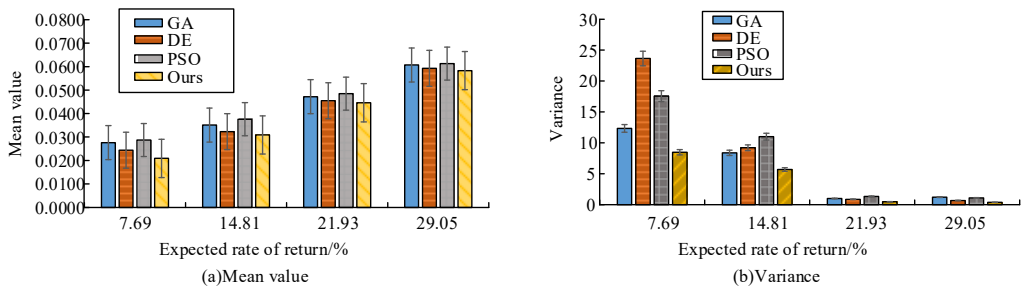

Fig. 2. Comparison of the results of different intelligent algorithms

From Fig. 2 (a), overall, the mean value increases continuously as the expected return rate increases. Among them, the mean value of the research algorithm is 0.0209,0.0309,0.0446 and 0.0583, which are smaller compared with the comparison algorithm. From Fig. 2 (b), the variance of the research algorithm is $8.4798E^{-05}$, $5.6818E^{-05}$, $4.7613E^{-06}$, $3.7313E^{-06}$, is smaller compared to the contrast algorithm. The above results show that the portfolio obtained using the research algorithm bears less risk for a given return. To further verify the effectiveness of the proposed method, the study selected 20 stocks will randomly generate two portfolios, each portfolio containing 10 stocks. The maximum, the minimum, the mean and the standard deviation of the two portfolios at different levels of risk yields are obtained, as shown in Fig. 3.

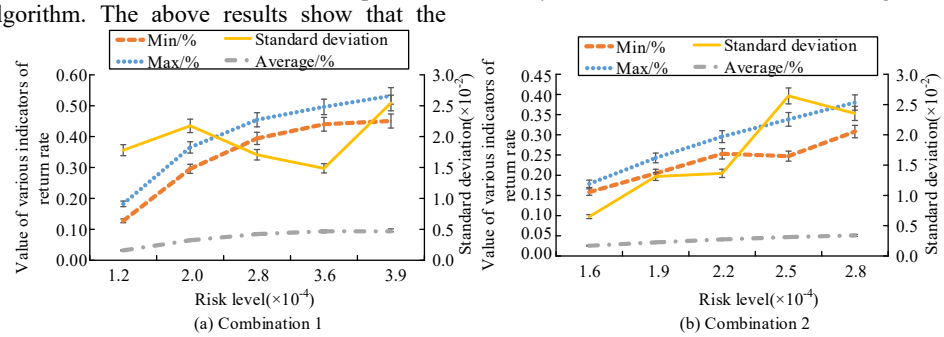

Fig. 3. 2 Index values of the returns of the investment portfolios at different risk levels

As can be seen from Fig. 3, in general, with the increase of the risk level, the maximum, minimum and average value of the yield rate all increase, but there is no obvious trend, indicating that with the increase of the risk level, the stability of the research algorithm remains basically unchanged. But as the level of risk increases, the maximum, minimum and yield of the average stabilize. From Fig. 3 (a), for portfolio 1, the risk level ranges from 1.210^{-4} was increased to 2.010^{-4}, the increase of the maximum, minimum and average value of the yield was 100.2%, 141.7% and 106.7%, respectively. While the risk level ranged from 3.610^{-4} was increased to 3.910^{-4} oclock, the maximum, minimum and average increase of the yield were 8.16%, 2.39% and 3.18% respectively. From Fig. 3 (b), for portfolio 2, the risk level ranges from 1.610^{-4} was increased to 1.910^{-4}, the increase of the maximum, minimum and average value of the yield was 26.43%, 22.63% and 24.96%, respectively. While the risk level ranged from 2.510^{-4} was increased to 2.810^{-4}, the maximum, minimum and average increase of the yield were 10.99%, 19.80% and 7.98%, respectively. The overall results show the reliability and effectiveness of solving the enterprise portfolio optimization problem. The optimal investment allocation of the 2 portfolios corresponding to different risk levels is shown in Table II.

TABLE II. RESULTS OF OPTIMAL INVESTMENT ALLOCATION CORRESPONDING TO DIFFERENT RISK LEVELS

Investment portfolio		Combination 1					Combination 2					
Risk level R (10^{-4})		1.2	2.0	2.8	3.6	3.9	1.6	1.9	2.2	2.5		
Actual risk R (10^{-4})		1.2	2.0	2.8	3.6	3.9	1.6	1.9	2.2	2.5		
Rate of return P/%		0.1519	0.3008	0.4441	0.4768	0.5330	0.1702	0.2212	0.2451	0.3008		
Investment share	1	0.219	0.066	0.050	0.098	0.009	0.040	0.074	0.064	0.146	0.031	
	2	0.055	0.076	0.145	0.206	0.197	0.100	0.136	0.210	0.220	0.202	
	3	0.235	0.213	0.224	0.071	0.209	0.123	0.015	0.114	0.130	0.025	
	4	0.019	0.052	0.042	0.073	0.004	0.053	0.039	0.036	0.052	0.015	
	5	0.100	0.156	0.210	0.281	0.166	0.099	0.034	0.075	0.078	0.187	
	6	0.049	0.005	0.005	0.000	0.041	0.176	0.119	0.100	0.062	0.006	
	7	0.088	0.152	0.191	0.145	0.229	0.039	0.087	0.001	0.018	0.032	
	8	0.127	0.107	0.044	0.017	0.016	0.013	0.050	0.085	0.020	0.134	

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AS shows in Table II, the yield increases. The investment share of stocks with lower average return tends to be 0 with the increase of risk level, but the investment share of stocks with higher average return rate. With the increase of the risk level, it slowly tends to 1, which shows that the algorithm for solving the enterprise portfolio optimization problem proposed by the research can effectively avoid risks and obtain high returns. When the risk level was changed from 1.210^{-4} was gradually increased to 2.010^{-4}, 2.8×10^{-4}, 3.6×10^{-4}, 3.9×10^{-4}, the return rate of portfolio 1 also increased from 0.1519% to 0.3008%, 0.4441%, 0.4768% and 0.5330%, indicating that the greater the risk, the higher the return obtained. The results justify the proposed algorithm.

V. CONCLUSION

With the increasingly complex social and economic environment, the traditional solution algorithm of enterprise portfolio optimization model can no longer meet the practical needs. In view of the current problems existing in portfolio optimization, the PSO and QAOA algorithm are studied to improve the generation mode of the initial population quantum coding, and propose a hybrid algorithm suitable for solving the mathematical model of enterprise portfolio optimization. The results show that the average value of this method is below 0.06, which is smaller compared with the comparison algorithm, and the obtained portfolio bears less risk. The stability of the study algorithm remains largely constant as the risk level increases. Risk level ranged from 2.510^{-4} was increased to 2.810^{-4} hours, the increase of the maximum, minimum and average value of the yield was 10.99%, 19.80% and 7.98% respectively, and the increase tends to be stable. The results show that the research algorithm is reliable and reasonable in solving the enterprise portfolio optimization problem. However, the algorithm proposed in this study is only applied in the enterprise portfolio optimization model, and will be further studied in the future to apply the algorithm to other economic models to improve the application scope of the algorithm.

REFERENCES

[1] Gunjan A, Bhattacharyya S. A brief review of portfolio optimization techniques. Artificial Intelligence Review, 2023, 56(5): 3847-3886.

[2] Chen Y, Zhao X, Yuan J. Swarm intelligence algorithms for portfolio optimization problems: Overview and recent advances. Mobile Information Systems, 2022, 2022(1): 4241049.

[3] Das A, Chaudhuri T, Roy S S, et al. Selection of appropriate portfolio optimization strategy. Theoretical and Applied Computational Intelligence, 2023, 1(1): 58-81.

[4] Erwin K, Engelbrecht A. Meta-heuristics for portfolio optimization. Soft Computing, 2023, 27(24): 19045-19073.

[5] Naik M J, Albuquerque A L. Hybrid optimization search-based ensemble model for portfolio optimization and return prediction in business investment. Progress in Artificial Intelligence, 2022, 11(4): 315-331.

[6] Brandhofer S, Braun D, Dehn V, et al. Benchmarking the performance of portfolio optimization with QAOA. Quantum Information Processing, 2022, 22(1): 25.

[7] Hegade N N, Chandarana P, Paul K, et al. Portfolio optimization with digitized counterdiabatic quantum algorithms. Physical Review Research, 2022, 4(4): 043204.

[8] Aguilera E, de Jong J, Phillipson F, et al. Multi-Objective Portfolio Optimization Using a Quantum Annealer. Mathematics, 2024, 12(9): 1291.

[9] Qu D, Matwiejew E, Wang K, et al. Experimental implementation of quantum-walk-based ortfolio optimization. Quantum Science and Technology, 2024, 9(2): 025014.

[10] Hamdi A, Karimi A, Mehrdoust F, et al. Portfolio selection problem using CVaR risk measures equipped with DEA, PSO, and ICA algorithms. Mathematics, 2022, 10(15): 2808.

[11] Erwin K, Engelbrecht A. Multi-guide set-based particle swarm optimization for multi-objective portfolio optimization. Algorithms, 2023, 16(2): 62.

[12] Shahid M, Ashraf Z, Shamim M, et al. Solving constrained portfolio optimization model using stochastic fractal search approach. International Journal of Intelligent Computing and Cybernetics, 2022, 16(2): 223-249.

[13] Naik M J, Albuquerque A L. Hybrid optimization search-based ensemble model for portfolio optimization and return prediction in business investment. Progress in Artificial Intelligence, 2022, 11(4): 315-331.

[14] Vaziri J, Farid D, Nazemi Ardakani M, et al. A time-varying stock portfolio selection model based on optimized PSO-BiLSTM and multi-objective mathematical programming under budget constraints. Neural Computing and Applications, 2023, 35(25): 18445-18470.

[15] Song Y, Liu Y, Chen H, et al. A multi-strategy adaptive particle swarm optimization algorithm for solving optimization problem. Electronics, 2023, 12(3): 491.

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